

μ-GIL: A Spatial Abstraction Learner

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Abstract

The Piagetian Modeler cognitive architecture marks an inflection point in field of artificial intelligence because it defines a new paradigm of constructivist neurosymbolic systems that autonomously construct or revise mental models of actual and virtual worlds solely through perception. These cognitive systems aim to achieve situational awareness, emotional regulation, individual autonomy and development, and a measurable degree of intelligence via cognitive system design patterns, concurrent stigmergic mechanisms, and a shared persistent memory. The Generally Intelligent Learner (GIL) was the first complete instantiation of this architecture. We present μ-GIL, a parsimonious working instance of the architecture specifically built to navigate the ARC-AGI-3 spatial reasoning corpus. Both cognitive systems faithfully implement a perception pipeline and Piagetian epistemology, however, μ-GIL replaces GIL's designed spatial ontology, concurrent solver system, formal reasoning engine, and developmental trajectory with an empirical, streamlined system. The core mechanisms, correspondences to biological theories, and results from initial testing against the ARC competition server are described herein.

Keywords μ-GIL, GIL, ARC-AGI-3, NAPOT, neural proposition, Piagetian Modeler, Premise language.

1. Introduction

A cognitive architecture is a blueprint – a specification of elements, their relationships, and interactions – for a system that navigates a real or virtual world, by synthesizing distinctions correlating to things in the world, forming logical, physical, and emotional responses to perceived or imagined situations, and achieving objectives by taking action in the world. A cognitive system is an implementation of a cognitive architecture. Cognitive systems are comprised of working software running on a hardware platform. The Piagetian Modeler cognitive architecture¹ (Figures 5, 6) is designed to implement aspects of human cognitive development using cognitive system design patterns which operate stigmergically on a shared memory.

1.1 Background

The Piagetian Modeler has three defining properties. First, perception is active and constructive: sensory input is transformed through a pipeline of detectors into a variable length trait vector—a structured representation of the perception. The trait vector is compared to stored experience. When a match is not found (i.e., the set of traits in the vector does not correspond an existing set of traits stored as “cases” within some threshold tolerance), the experience is assimilated: meaning that a new identifier called a “reifier” encompassing all the traits is minted, a new case containing the traits and reifier is stored, and the mind's ontology grows. When a match is found, the experience is accommodated, meaning that the existing memory is reinforced and activation spreads throughout the ontology.. No new representation is created. This is the central hinge. No probabilistic threshold, no loss function, no gradient – a categorical decision made by the Matching mechanism, with consequences rippling through the entire system.

Second, the world model is acquired incrementally, from observed consequences of actions, not given a priori as training data. Every action the system takes in the world produces a proprioceptive feedback that is perceived, compared to prior predictions, and used to update the model. A confirmed prediction reinforces the model; a failed prediction triggers compensation—the model is corrected at exactly the point of surprise. Over time, the system builds a history of events based on the actions taken, and uses those events to form episodes and stories, lessons and hypotheses. All of which are used o imagine (i.e., simulate) possible future outcomes. The transition from taking actions to dreaming is the transition from sensorimotor to operational intelligence—Piaget's developmental trajectory².

Third, the memory substrate is shared across all components. Perception, coordination, reflection, and consolidation all read from and write to the same knowledge base. A knowledge unit minted by perception is the same unit the simulator dreams with, the same unit a coordinator scores, and the same unit persistence serializes. There is no separate training data, no separate inference model – the mind's working memory is its long-term memory, partitioned into areas (perceptual, ontological, associative, activative) but unified in substrate.

GIL and μ -GIL are both cognitive systems built on this architecture. They share the Piagetian Modeler epistemology, the active perception pipeline, the learned world model, and the shared memory substrate. They differ in scope, mechanism inventory, and implementation strategy. Differences this paper documents in detail.

2. GIL: A Generally Intelligent Learner

GIL^{3, 14} is a cognitive system coded in the Premise Programming Language⁴ (version 3.3). GIL is the reference implementation of the Piagetian Modeler architecture. The full GIL system comprises approximately 50 source files totaling over 20,000 lines of Premise language code (see Figure 9). The system is organized around four concurrent processes (observation, coordination, reflection, and consolidation) operating over a shared memory, the Totality, that has six memory areas (control, ontological, perceptual, associative, activative, and imaginative).

Observation processes implment three observation design patterns implementing the perception pipeline: a multi-modal Perceiver mechanism receives percepts from world interfaces (called Psyches), forwards them to Detectors that extract traits, then forwards the traits to a Matcher mechanism that

compares the traits stored experiences (cases). If not matched within some threshold, a Lexer mechanism mints a unique 128-bit reifier in ontological memory (as a scheme or neural proposition¹³) representing the percept, and a Storer mechanism places the reifier as a case into associative memory. Finally, an Activator mechanism stamps the found (or created) reifier and its traits as active as of the indicated moment (See Figure 1). The Observation detectors include an Imager mechanism for image schema formation based on detected objects. Spatial representation in GIL follows a seven-level hierarchy: Scene, Venue, Locale, Place, Area, Region, Map. Three dimensional spatial coordinates bind objects to positions. Image schemas serve as spatial templates. A Navigator performs BFS pathfinding over the spatial graph, and a Locator/Mapper pair constructs and maintains the hierarchy.

Coordination patterns implement goal-directed behavior through a motivational urge Ameliorator mechanism that converts needs to objectives, ten concurrent problem solvers propose candidate actions, a Deliberator mechanism selects actions based on their relevance to outstanding objectives, a Reactor mechanism selects actions based on their reliability, and a Correlator/Regulator/Terminator chain grades action consequences. Reasoning is handled by 136 operators across 7 reasoning strategies, implementing formal inference over the ontological memory. Structuring mechanisms build heterarchic abstractions through clustering, a Hebbian image-schema⁵ formation mechanism (the Skemer) uses co-activated reifiers, and narrating mechanisms produce Stories and Theories (Kuhnian paradigms that organize knowledge⁶).

Reflection contains sub-patterns for motivation, simulation (daydreaming), failure compensation, emotional coping, hypothesis discovery and experimentation, imitation, and play – all developmentally gated to activate at appropriate developmental stages. Consolidation processes handle compressing (chunking), automating (skill formation), forgetting (decay-based pruning), and recycling (resource reuse).

3. μ -GIL: A Parsimonious Spatial Learner

μ -GIL is a second implementation of the Piagetian Modeler architecture. The purpose was twofold: to validate GIL's perception pipeline against the ARC-AGI-3 testbed^{7, 8, 9}, and to discover which coordination and reflection mechanisms a working spatial abstraction learner actually needs. The implementation comprises approximately 30 source files – 13 mechanism theories, 2 psyche theories, a totality memory module, an imagination module, a portal renderer, configuration, boot sequence, and an orchestration stack (bridge, orchestrator, PTY driver, grind loop, ablation scripts) – totaling roughly 8,000 lines of Premise and 1,200 lines of Python code (see Figure 10).

3.1 What μ -GIL shares with GIL

The perception pipeline is implemented faithfully. A Psyche world interface mechanism (named Camera for the the ARC-AGI-3 world (see Figure 4)) sends PERCEPT tuples to the perceiver mechanism (see Table 1). Detector mechanisms extract traits, the Matcher mechanism tests for similarity (Jaccard similarity at a configurable threshold), the Lexer mechanism mints reifiers and creates Schemes on miss, the Storer mechanism creates Cases, and the Activator stamps recency. The Scener mechanism captures object arrangements as Scenes that imagination later remixes. Every observation logs an audit trail trace at each pipeline stage – thereby making perception scrutable.

The Piagetian epistemology is intact. Assimilation (a Matcher miss mints a new scheme) and accommodation (a Matcher hit reinforces the existing case) are the only two paths through the pipeline. The Correlator grades every action's consequence by comparing the action result against the observed outcome – confirming or invalidating the prediction – and triggering compensatory corrections in the case of misprediction.

3.2 What μ -GIL replaces

The spatial representation is entirely μ -GIL's own invention. Where GIL specifies a seven-level designed hierarchy with spot binding and image schemas, μ -GIL builds an empirical arrangement graph from observed transitions. Connected-component labeling (a two-pass algorithm) extracts regions from grid frames. A controllable-color whitelist – colors whose regions have been observed to move under actions, as opposed to HUD elements that vanish and appear – filters these regions into an Arrangement: a multiset of "color@x,y" keys that serves as a position fingerprint. The canonical arrangement ID (the Case context of the first sighting) becomes the key for all downstream structures: the PairTried frontier ledger, the chain-walk edge graph, and the ClusterShift attribute-transition table.

The μ -GIL coordination architecture replaces GIL's ten concurrent solvers with a single-threaded band auction. Approximately 15 scoring bands compete deterministically (see table 2). The highest score wins; first-max breaks ties. GIL's 136 reasoning operations are absent. μ -GIL has no formal reasoning engine. Where GIL would derive conclusions through inference chains, μ -GIL relies entirely on learned transitions (what an action did from a context) and dreamed predictions (fork the current scene, apply the scheme's operators, return predicted keys and traits). The accommodation mechanism exists – a surprised prediction overwrites the original – but it operates at the level of a single overwrite per misprediction.

GIL's Reflection components (motivation, simulation, compensation, coping, discovery, imitation, play) are reduced to simulation (i.e., daydreaming) alone. GIL's Consolidation component (compressing, automating, forgetting, recycling) are absent, except for a stale-cluster cull (24 consecutive unmatched taught frames delete the row) and anchor lifecycle resets on scene changes. GIL's developmental tier gating, narration, and theorizing are all absent.

3.3 What is unique to μ -GIL

Several mechanisms were discovered empirically that GIL's specification does not anticipate. These emerged from the iterative cycle of running μ -GIL against the live competition, diagnosing failures from the learning curves and deliberation audit trails, building the mechanism that addresses each failure, and verifying it with a unit demo before the next live run.

The relational state layer (Clusters and ClusterShifts) discovers board patterns without any game-specific knowledge. Proximity groups of small same-color regions are tracked across frames and summarized by normalized pattern tokens: binarize the cluster's bounding box by its color, crop to

the figure, downscale by the per-axis GCD of run lengths, serialize as "WxH:bits". This normalization makes patterns comparable across positions and across render scales – a glyph drawn in 2-by-2 blocks yields the same token as the same glyph drawn in single cells. A mutable cluster (its token shifts under actions) whose alphabet of shown patterns contains a static cluster's token adopts that token as its anchor: "this indicator can show what that plate shows". The match predicate fires when the indicator currently shows its anchor. No coordinates, no color codes, no glyph patterns are assumed – everything is discovered from region dynamics.

The chain-walk is a bounded best first search (BFS) over the learned arrangement graph (PairTried edges, observed at settle time) that propagates terminal value back to each candidate's first action. Anchored-cluster tokens ride along each leg via ClusterShift rows, so the walk knows where a match closes, holds, or breaks. Terminal value is discounted per leg below the one-step bands: chain-progress (335/330/325), chain-match (325/320/315), chain-carry (244/243/242), chain-frontier (108/106/104). The walk addresses two diagnosed one-step-ceiling failures: the pocket escape (every one-step future is known and burned, but a frontier sits three legs away) and the match carry (a relational match must be valued across the distance between the indicator and the plate).

The muse channel supplies external direction without supplying belief. A routing mechanism reads the game's frame files and writes hypothesis files containing ordered waypoint lists. The μ -GIL follows the current waypoint only through its own machinery: a candidate whose dreamed future approaches the waypoint scores muse-directed 248 – above explorer (245) so a committed multi-leg excursion sustains through fresh territory, below every trusted band (300 and above) so the prospect never outranks evidence. Hypotheses must earn their standing: per-leg patience (20 tries per waypoint before retirement), level gating (a level-2 route stays silent during level-1 play), and per-life reset (a world teleport restarts the route from waypoint 1). The ablation contract is structural: empty hypothesis directory means zero rows, zero invocations, zero behavior change.

The budget gate learns the world's action budget from visible teleports. LS20 level 2 grants approximately 21 actions per life, then teleports the piece home and reverts the key display. The mind counts settles between teleports, learns the minimum gap as the budget, and vetoes any carry whose mover cannot reach the target plate within the remaining budget. Active only after two consistent teleport gaps are observed – a single gap is treated as a fluke.

Evidence confidence replaces latest-observation-overwrites with majority-based edge recording. Agreement with the recorded edge reinforces it; disagreement is counted, and only a new majority flips the edge. A single fluke observation – the exact failure mode that built the level-2 sink in the live run – cannot overwrite 26 consistent observations. Question-the-map addresses the complementary failure: when the chain-walk proves the map has no way out (it ran and returned no verdict at any depth), the mind re-tries the least-recently-settled pair of the current arrangement, applying accommodation to the frontier ledger itself.

4. Spatial Abstractions

A spatial abstraction is a representation that captures the structure of a spatial arrangement at a level above raw coordinates. The raw grid has cells with colors at positions. A spatial abstraction groups cells into regions, groups regions into arrangements, tracks how arrangements change under actions, discovers which arrangements contain goal-relevant patterns, and plans multi-step routes through the space of arrangements toward those goals. Each level of abstraction removes detail and adds structure: cells become regions, regions become arrangements, arrangements become nodes in a navigable graph, and the graph becomes a planning substrate.

4.1 GIL's Designed Spatial Ontology

GIL starts with a seven level spatial hierarchy (See Figure 2): with Scenes at the bottom and one or more maps at the top. Scenes (the current perceptual field), Venues (the navigable space containing the scene), Locales (a bounded subspace with a landmark- indoors, outdoors, open, enclosed), Place (a recognizable location within a locale), Area (a functional grouping of places), Region (a perceptually cohesive grouping of areas), and Maps (the global environment). Each venue in GIL is a 3D canvas that is perceived, recalled, or imagined. This hierarchy is constructed and maintained by two mechanisms. The Locator (builds the hierarchy bottom-up from perceived spatial relations (containment, adjacency, above/below, near/far) associatively matched to known venues (see Figure 3), and the Mapper accumulates spatial groupings top-down as new groupings are encountered when routes are traversed.

Spots are 3D spatial coordinates binding objects to scenes within a venue. Ten image schemas⁵ serve as spatial templates that the Skemer (in associating.pattern) forms through Hebbian co-activation of scheme components. A Navigator performs BFS pathfinding over the spatial graph, and routes are first-class objects linking venues through spot waypoints.

In GIL the spatial map containers exist a-priori, before they are experienced. The hierarchy levels, the image schemas, the Locator/Mapper mechanisms, and the Navigator's pathfinding algorithm are all specified in the architecture. Experience fills the hierarchy – places are discovered, routes are learned – but the structural grammar is innate. This is a design choice: Piaget argued that spatial representation develops through a sequence of stages (topological → projective → Euclidean), and GIL's tier-gated developmental mechanism controls which levels of the hierarchy become active at which developmental stage.

4.2 μ -GIL's Empirical Spatial Ontology

μ -GIL has no spatial hierarchy, no image schemas, no Locator, no Mapper, no Navigator, and no prior spatial grammar. Its spatial representation is learned entirely from observed transitions, built from four layers of abstraction that emerge from the perception pipeline and the Coordinator's experience record.

The first layer is regions. A two-pass connected-component labeling algorithm groups contiguous same-color cells into regions, each characterized by a trait set: color, bounding box (x, y, width, height), cell count, and shape class. This runs every frame as part of the grid-state detector. The output is a flat list of ~24 region traits per frame – no hierarchy, no containment, no adjacency. The regions are the spatial primitives, analogous to GIL's Region level but without any of the levels above it.

The second layer is arrangements. A controllable-color whitelist – colors whose regions have been observed to move under actions, as opposed to HUD elements that vanish and appear regardless – filters the region list into a position fingerprint: a multiset of 'color@x,y' keys. The canonical arrangement ID (the Case context of the first sighting) becomes the key for all downstream structures. An Arrangement is 'where the controllable pieces are' – a spatial state that recurs when the same pieces occupy the same positions. This is μ -GIL's substitute for GIL's Place: not a named location in a hierarchy, but an empirical fingerprint that recurs.

The third layer is the edge graph. Each time μ -GIL chooses an action from a known arrangement and arrives at another arrangement, the PairTried ledger records the edge: (arrangement, action) $\rightarrow$ destination arrangement, with a settle-order stamp and majority-based evidence confidence. Over time, this builds a directed graph where nodes are arrangements and edges are actions. This is μ -GIL's substitute for GIL's Map: not a pre-specified graph maintained by a Mapper, but an empirical graph that grows edge by edge from observed transitions. The chain-walk (a bounded BFS over this graph, depth ≤ 7 , for approximately 64 visited nodes) is μ -GIL's substitute for GIL's Navigator.

The fourth layer is relational state. Clusters – proximity groups of small same-color regions – are tracked across frames and summarized by normalized pattern tokens: binarize the cluster's bounding box by its color, crop to the figure, downscale by the per-axis GCD of run lengths, and serialize as 'WxH:bits'. A mutable cluster whose token shifts under actions (an indicator) that shows a static cluster's token (a plate) adopts that token as its anchor. The match predicate fires when the indicator currently shows its anchor. This relational discovery – 'this display can show what that door shows' – is μ -GIL's substitute for GIL's image schemas: not an innate spatial template, but a pattern discovered from dynamics.

4.3 What Learning Means

The progression from zero spatial knowledge to a navigable, predictable, relationally-annotated mental model is visible in the learning curves (Chart 1). The frontier burn-down tracks untried (arrangement, action) pairs – rising when new arrangements are discovered, falling as pairs are tried. The assimilation ratio tracks the share of arrivals at previously-seen arrangements. The exploration rate tracks the share of decisions that try untried pairs. In the first run, the frontier plateaus at 30 – the mind oscillates in known territory without exhausting its options. By the seventh run after mechanism development, the frontier burns to 22, the assimilation ratio climbs faster, and the exploration rate drops to zero – the mind has learned enough of the spatial structure to exploit it.

The cost-per-win data (Chart 4) shows the aggregate consequence: from ~640 actions per level-1 win early in the campaign to ~125 actions per win at the end, against a theoretical floor of 15 actions (the scripted probe that already knows the route). The gap between 125 and 15 is the

exploration cost – the price the mind pays for building its spatial abstractions from scratch rather than receiving them as given.

4.4 Neural Rendering

The predict-compare-update loop at the core of μ -GIL has an independent parallel in biological vision theory. Blumberg¹⁰ proposes that biological vision is not passive image classification but a recurrent sensorimotor process in which action changes evidence, predicted consequences are compared to observed refference (when an organism's movements cause changes in sensory perception, e.g., foveation or turning one's head), and the perceptual mismatch updates the model. He argues that a useful theory of visual learning must explain how an agent separates self-caused from externally caused changes in perception, how an agent maintains state across its own movements, how an agent predicts consequences, and how an agent revises its model when prediction and refference disagree. This is the same epistemic structure that governs μ -GIL's core loop.

The mechanism Blumberg proposes for this process he calls “Neural Rendering”: a causal state update in which distributed inputs are transformed and re-expressed to downstream receivers. The term rendering does not imply a picture presented to an inner spectator – it names the operation by which patterned input is received, integrated with recurrent expectations and motor context, and re-expressed as an updated body-and-world state that conditions the next perception and action. Blumberg formalizes the embodied-vision causal sequence mathematically (ibid., Section 5). For world state, body state, and selected action, a forward model predicts both world and body consequences:

$$\hat{y}_{\text{world}}, \hat{y}_{\text{body}} = F(s_{\text{world}}, s_{\text{body}}, a) \quad (\text{i})$$

where $\hat{y}_{\text{world}}$ and $\hat{y}_{\text{body}}$ are the predicted world and body states, F is the learned forward model, s_{world} is the current world state, s_{body} is the current body state, and a is the selected action.

The post-action observation is encoded into a structured residual – the difference between predicted and observed:

$$\delta = \text{encode}(y) - \hat{y} \quad (\text{ii})$$

where δ is the structured residual, y is the actual post-action observation (the refferent sensory return), encode is the perceptual encoding function, and $\hat{y}$ is the predicted observation from equation (i).

Blumberg insists this residual must not be collapsed to a single scalar but decomposed into perceptual, object, relational, transition, body, rule, and goal components. The state update is then governed by an empirically specified gate:

$$s_{t+1} = G(s_t, \delta, a) \quad (\text{iii})$$

where s_{t+1} is the updated state, s_t is the current state, G is an empirically specified gate function that determines which components of the residual δ update which state variables, and a is the action that produced the residual.

Blumberg further separates conditional surprise from Bayesian information gain – quantities he notes are routinely conflated (ibid., Section 8.1). Conditional surprise measures the improbability of an observation under the predictive model:

$$S(y_t | y_{<t}, a_{<t}) = -\log p(y_t | y_{<t}, a_{<t}) \quad (\text{iv})$$

where S is the conditional surprise, y_t is the observation at time t , $y_{<t}$ is the history of all observations before time t , $a_{<t}$ is the history of all actions before time t , and p is the probability assigned by the predictive model. Information gain measures belief revision about a latent state:

$$I(z; y_t | y_{<t}, a_{<t}) = D_{KL}[p(z | y_{\leq t}, a_{<t}) \parallel p(z | y_{<t}, a_{<t})] \quad (\text{v})$$

where I is the Bayesian information gain, z is the latent state variable, D_{KL} is the Kullback-Leibler divergence, $p(z | y_{\leq t}, a_{<t})$ is the posterior belief about z after observing y_t , and $p(z | y_{<t}, a_{<t})$ is the prior belief about z before observing y_t .

Neither quantity is evidence that the signal causally changed a downstream receiver. Perceptual clarity, semantic relevance, and causal effectiveness require separate endpoints.

The correspondence between Blumberg's formalism and μ -GIL's implementation is specific. In μ -GIL, the forward model F (equation i) is the TransitionScheme row: action + context $\rightarrow$ predicted moved/vanished/appeared. The structured residual δ (equation ii) is the Correlator's symmetric difference over predicted and observed arrangement keys – it lists the specific regions at positions that the dream missed, categorized as missing (predicted but not observed) or extra (observed but not predicted). The gate G (equation iii) is the Correlator/Regulator/Terminator chain that routes confirmed predictions to reinforcement and surprised predictions to accommodation – the model overwrite at the point of misprediction. The action copy is the scheme lookup; the forward model is the TransitionScheme row; the reafference comparison is the dream-vs-actual diff.

Blumberg's corollary-discharge model describes the same chain: a selected motor command is copied into a forward model, the model predicts sensory consequences, movement changes the sensory return, and the observed return is compared with the prediction – with a mismatch updating the pose estimate, action calibration, or world model. Blumberg's separation of world/object state from body

state – made operational through independent measurement endpoints for pose, agency, ownership, and action possibility – has a functional counterpart in μ -GIL's controllable-color whitelist, which learns to separate the agent's piece (its body in the grid world) from HUD churn (world state the agent cannot affect).

4.5 Scheme Activation

The parallel extends beyond the predict-compare-update loop to the mechanism of activation itself. Blumberg's Neural Array Projection Oscillation Tomography (NAPOT) (See Figure 7, 8) describes a fundamental computational unit such as a neuron, where multiple partial projections converge at a receiving unit (dendrites connecting to a soma), the receiver transforms the combined input, and the result is re-expressed to downstream receivers (the soma arborizes to the axons) – what Blumberg calls receive-transform-re-express. GIL's scheme activation follows the same three-phase operation: afferent monads (a scheme's arguments – the perceptual or associative inputs that bear on it) coalesce activation at the scheme node; the scheme integrates (its activation level crosses the threshold that makes it currently relevant); and efferent monads (the scheme's referents) receive dispersed activation. GIL's activating pattern implements spreading activation – activation levels spread through associative memory with decay rates and temporal windows that govern when afferent activation becomes effective at the receiving scheme. GIL activates schemes by time stamping them with the current moment, whereas Blumberg advocates an oscillatory phasing.

In μ -GIL, the Activator itself is a binary mark of relevance – the temporal dynamics of GIL's spreading activation are absent. However, the convergence-integration-projection topology appears in several other mechanisms.

These parallels operate at the level of shared Piagetian epistemology and shared computational topology. Blumberg's specific contributions – phase-structured laminar interactions in the Gamma Consideration Sandwich, dendritic radial-basis fields, the oscillatory reconstruction claim in NAPOT – describe the neural substrate that implements these operations in biological tissue. μ -GIL implements the same operations in symbolic Premise code over a flat knowledge base. The connection is architectural: both systems instantiate the Piagetian Modeler's core epistemic loop and its convergence-integration-projection activation cycle, one in proposed biological hardware and the other in working software against a live task.

Blumberg formalizes the NAPOT receive-transform-re-express operation at a single receiving unit (ibid., Section 6). The standardized candidate drive at receiving site j is:

$$d_j = \sum_i K(\mathbf{x}_j, \mathbf{x}_i) \cdot w_i \cdot u_i \cdot \psi(\tau_i - \delta_i) - I_j \quad (vi)$$

where d_j is the candidate drive at receiving site j , K is a localized kernel encoding spatial selectivity between sites j and i , $\mathbf{x}_j$ and $\mathbf{x}_i$ are the spatial positions of the receiving and sending sites, w_i is the learned efficacy (synaptic weight) of input i , u_i is the incoming afferent drive (the signal arriving from

source i), ψ is a bounded receiver-window function that gates temporal effectiveness, τ_i is the arrival timing of input i , δ_i is an independently constrained transmission delay for input i , and I_j is the inhibitory contribution at site j .

The event-related field – the phasic signal that drives downstream re-expression – is then the departure from a tonic (expected) baseline:

$$\tilde{d}_j = d_j - \langle d_j \rangle_{\text{tonic}} \quad (\text{vii})$$

where $\tilde{d}_j$ is the event-related field at site j , d_j is the candidate drive from equation (vi), and $\langle d_j \rangle_{\text{tonic}}$ is the tonic expectation — the training-defined or independently localized baseline drive at that site. The event-related field isolates the phasic departure: what is new, relative to what was expected.

Each component of equation (vi) has a specific counterpart in GIL's activating pattern. The incoming drive u corresponds to the Pulses arriving from afferent monads (argument schemes) via the Observer agent's upward spread through associative memory. The receiver-window function ψ corresponds to GIL's EnabledSince/EnabledUntil temporal window on the Activation record – the signal is effective only within a bounded interval, enforced by Cronos, which expires activations whose window has elapsed at its @m50 cycle. The inhibitory contribution I corresponds to GIL's impede mechanism: negative Pulses with their own temporal window (ImpededSince/ImpededUntil), separately gated by Cronos. The localized kernel K – not every input is equally effective – corresponds to GIL's ArgumentQuorum: the scheme does not integrate until a threshold number of its specific afferent monads have converged, providing selectivity over which subset of inputs triggers the integration.

The event-related field (equation vii) – the phasic departure from tonic expectation – has a structural counterpart in the Observer/Anticipator asymmetry. Observer spreads activation upward (argument → reifier → referent) in the Observed reality at 1 moment (0.00315 seconds); Anticipator spreads activation downward (referent → reifier → argument) in the Expected reality, also at 1 moment. Both operate at the same cycle rate in opposite directions. The difference between Observed activation and Expected activation arriving at a scheme is the phasic residual – the surprise that triggers accommodation when prediction and observation disagree.

GIL's Realities enumeration (Observed, Expected, Recalled, Imagined) further parallels NAPOT's characterization of the same receive-transform-re-express operation running in different modes. Each reality is served by its own agent at a different cycle rate: Observer at 1 moment, Anticipator at 1 moment, Reminder at 5 moments, Dreamer at 10 moments – with different propagation directions (upward, downward, radial, upward respectively). The temporal stratification of these agents – fast sensory observation and expectation, slower recall, slowest imagination – mirrors Blumberg's proposal that different frequency bands serve different functional roles in the cortical laminar architecture.

μ -GIL's Activator is binary – no temporal dynamics, no quorum, no spreading. That does not have NAPOT correspondence. But the Matcher does, and so do several other μ -GIL mechanisms, at a different level.

The Matcher's Jaccard scan is a convergence-integration-projection operation. Multiple traits from the current percept (partial projections from different feature dimensions – color, position, size, shape) converge at each stored Case. The Jaccard overlap computation is the integration – how much of the incoming trait set aligns with the stored trait set. The 85% threshold is a quorum gate – the Case does not activate (get recognized) unless enough traits converge. The best-above-threshold selection is the re-expression – the winning Case becomes the recognized scheme that flows downstream to the Activator, Scener, and Coordinator.

That is NAPOT's receive-transform-re-express, with the Jaccard threshold playing the role of ArgumentQuorum. It is not temporal – there is no decay window, no cycle rate, no phase – but the topology is right: multiple afferent signals converge at a receiver, a threshold determines whether integration succeeds, and the result is re-expressed downstream.

The band auction in the Coordinator is another instance. Multiple reasons from different solvers (kace, dream-progress, dream-match, reactor, explorer, muse-directed) converge at each candidate action, each contributing a score. The highest score wins – integration with a winner-take-all gate rather than a quorum gate. The chosen action is re-expressed as the decision that flows to grid-act. Multiple partial evaluations converging at a receiving node, integrated by competition, re-expressed as the decision.

The chain-walk is a third instance. Value propagates backward through the arrangement graph: terminal value at depth N is discounted and projected back to the first-leg action at depth 1. Multiple legs contribute partial evaluations (chain-progress, chain-match, chain-carry, chain-frontier) that converge at the root. This is literally spreading activation through a learned graph – the closest μ -GIL gets to GIL's spreading activation, though it is bounded ($\text{depth} \leq 7$) and runs once per decision rather than continuously.

μ -GIL thus has NAPOT correspondence at the topological level – convergence-integration-projection operations with threshold or winner-take-all gates – but not at the temporal level. The WHAT and WHERE of NAPOT are present; the WHEN is missing. Implementing GIL's full activating.pattern, with its temporally-windowed convergence, quorum-gated integration, bidirectional Observer/Anticipator spreading, and multi-reality agents, would add the temporal dimension that closes the gap between μ -GIL's topological correspondence and GIL's fuller alignment with the dynamics Blumberg's NAPOT equations describe.

5. The Runtime Environment

μ -GIL runs on Premise 3.3.alpha, a pre-release runtime. Due to several restrictions on the alpha version all hot paths were inlined into single functions: the frame parser, the grid detector, the cluster-pass, the chain-walk, and the dream-transition are each one function call per frame. An 8-turn session (the budget ceiling for the grid-play loop) exhausts the budget and saves the memory; the orchestrator (play.py) chains sessions, and the bridge (bridge.py) handles all protocol detail μ -GIL

must never see – HTTPS, cookies, JSON, 64-by-64 frame flattening – communicating through a file protocol of plain-token frame files and action files.

6. Results

The run record comprises 47 ledger entries from July 2 to July 3, 2026: 1 replay, 1 scripted probe, and 45 live LS20 runs against the ARC-AGI-3 competition server. Each live run consists of up to 16 chained 8-turn sessions (approximately 129 actions maximum), with the Totality saved and restored across session boundaries.

The first 11 live runs (July 2–3) completed zero levels. The perception pipeline was operational – frames were perceived, Cases stored, Scenes captured – but the Coordinator lacked the frontier-pair rule, the Simulator's dream bands, and the relational state layer. Run 12 (July 3, timestamp 134449) was the first to complete level 1, following the introduction of the Simulator's dream-as-prediction mechanism and the frontier-pair exploration rule. From run 14 onward (July 3, after the relational state layer landed), level 1 was completed in 30 of the remaining 34 full runs – a win rate that stabilized above 80% within the first week and remained there through the end of the campaign.

The scripted probe (run 2, July 2) banked level 1 in 15 actions – a target the mind has not yet matched on its own machinery (typical mind-driven level-1 wins use 80–120 actions to explore, learn, and exploit). The gap between 15 and 80 is the cost of exploration: the probe knew the route; the mind must discover it.

Level 2 was never completed. The campaign against level 2 is documented in the project journals and the level-2 energy-budget analysis. The core obstacle is a 21-action energy budget that μ -GIL must learn from observed teleports: the winning route requires approximately 19 actions (spawn to cross to portal), and the budget gate, muse channel, and per-life route reset were all built to address specific diagnosed level-2 failures. Five muse probes tested the empirical corridor waypoint route; the furthest reached the cross (the key-display rotator) but did not enter the portal before the budget expired. The level-2 campaign remains an open frontier.

Charts 1 and 2 present the learning curves. Chart 3 shows the rolling level-1 win rate stabilizing above 80%. shows the level-attainment progression from zero to reliable. Chart 4 shows learning efficiency: cumulative investment versus return and cost per level-1 win declining from ~640 actions/win to ~125 actions/win as the ken matures – against a theoretical floor of 15 actions (the scripted probe that already knows the route).

7. Discussion

The relationship between GIL and μ -GIL is one of two systems built on the same architecture. The Piagetian Modeler architecture – four components over a shared Totality, the assimilation–accommodation epistemic loop, is recognizably the same blueprint. The systems differ in which components are implemented, which mechanisms fill them, and which data structures represent space. GIL is the reference system; μ -GIL is a minimal working system that omits what the task does not demand and invents what the task does.

The closest alignment is in the perception pipeline. Both follow the same Table 1 sequence, and μ -GIL's implementation is faithful to the specification. The Matcher's Jaccard similarity threshold (85%, configurable) implements Piaget's assimilation criterion: experiences sufficiently similar to an existing scheme are absorbed into it; experiences sufficiently novel create new schemes. The trait-signature hash (a deterministic rolling hash mod 2^{31}) ensures that identity survives restarts – the same experience, detected again after a persistence round-trip, yields the same reifier. This content-addressed identity is what makes cross-session recall work by construction rather than by lookup.

The deepest divergence is in spatial representation. GIL's seven-level hierarchy (Scene through Map) with Spots binding and image schemas is a deductive approach: the spatial structure is constructed from perceived relations by rule-based Locator and Mapper mechanisms. μ -GIL's arrangement graph is an empirical approach: the spatial structure is learned from observed transitions, one edge at a time, with no prior spatial ontology. This is arguably the most significant finding: a system built on the Piagetian Modeler architecture can learn spatial structure from scratch, without a built-in spatial hierarchy, if its perception pipeline produces position-bearing traits and its coordination loop records which transitions lead where.

The coordination divergence – band auction versus concurrent solvers – is partly a matter of implementation constraint (the single-threaded Premise session cannot run concurrent agents) and partly a design choice. The band auction's total ordering (higher score wins, first-max breaks ties, rotation prevents persistent ties) is deterministic and fully auditable: every decision's score, basis, dream, and outcome are recorded on the Deliberation row and forwarded to the ARC API's reasoning field. The scoring bands themselves encode a theory of decision-making under uncertainty: direct progress evidence outranks simulated progress, which outranks reliable reflexes, which outranks untried exploration, which outranks modeled novelty. This hierarchy was not designed top-down but discovered iteratively – each new band was the fix for a diagnosed live failure.

What μ -GIL invented fills gaps that GIL does not address – not because they are omitted by oversight, but because they are the kind of mechanism that becomes visible only when a system confronts a real task. The arrangement graph, the cluster-based relational discovery, the evidence-confidence majority system, the budget gate, the muse channel, and question-the-map are all responses to specific, diagnosed failures in live play. They are documented in the project journals with the failure, the diagnosis, the mechanism, and the verification – the kind of engineering narrative that a specification cannot contain.

The absence of GIL's reasoning, reflection, consolidation, narration, and developmental gating in μ -GIL does not imply these are unnecessary. It implies they are unnecessary for level-1 of a specific ARC puzzle. A task requiring multi-step inference, emotional regulation, narrative memory, or developmental staging would expose their absence. μ -GIL's contribution is not a claim that the minimal system is sufficient – it is an empirical map of which architectural components suffice for spatial abstraction learning and which mechanisms the task demands that GIL does not yet provide.

8. Conclusions and Future Work

μ -GIL demonstrates that a minimal cognitive system built on the Piagetian Modeler architecture – faithfully implementing the perception pipeline, the assimilation–accommodation epistemology, while replacing the designed spatial ontology with a learned arrangement graph – can play ARC-AGI-3 live, win level 1 reliably from cold, and document its own learning in auditable decision records. The core system was built in two days, with subsequent campaign mechanisms developed over ten more, all running on a pre-release Premise runtime. μ -GIL discovered several mechanisms (relational state matching, chain-walk planning, energy-budget learning, imported-hypothesis channels) that extend GIL into territory it does not yet cover.

The open frontier is level 2, where the energy budget, the muse channel's waypoint routes, and the per-life route reset have been built and tested but not yet combined into a winning run. The level-2 campaign continues. The code, the run record, the learning curves, and the decision-level audit trails are the evidence.

Acknowledgements

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References

First Citation order

- [1] Miller, M. S. P. (2021). The Piagetian Modeler. In: Goertzel, B., Iklé, M., Potapov, A. (eds) Artificial General Intelligence. AGI 2021. Lecture Notes in Computer Science, vol 13154. Springer, Cham. DOI: 10.1007/978-3-030-93758-4_16.
- [2] ARC Prize Foundation (2026). ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence. arXiv:2603.24621.
- [3] Piaget, J. (1952). The Origins of Intelligence in Children. W. W. Norton and Company. DOI: 10.1037/11494-000.
- [4] Miller, M. S. P. (2026). Premise Language Reference Manual, version 3.389. SubThought Corporation.
- [5] Johnson, M. (1987). The Body in the Mind: The Bodily Basis of Meaning, Imagination, and Reason. University of Chicago Press.
- [6] Kuhn, T. S. (1962). The Structure of Scientific Revolutions. University of Chicago Press.
- [7] ARC Prize Foundation (2026). ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence. arXiv:2603.24621.
- [7] Miller, M. S. P. (2013). The Neural Proposition: Structures for Cognitive Systems. AAAI Spring Symposium Technical Report, pp. 44–50.
- [8] Chollet, F. (2019). On the Measure of Intelligence. arXiv:1911.01547.
- [9] Chollet, F., Knoop, M., Kamradt, G., and Landers, B. (2025). ARC-AGI-2: A New Challenge for Frontier AI Reasoning Systems. arXiv:2505.11831.
- [10] Blumberg, M. (2026). Embodied Neural Rendering in Self-Aware Networks: A Falsifiable Model of Vision, Sensorimotor Feedback, Body Maps, and the Gamma Consideration Sandwich. Version 1.1. Self Aware Networks Institute. DOI: 10.5281/zenodo.21331180.
- [11] Miller, M. S. P. (2018). Building Minds with Patterns. Independently published. ISBN 978-1-980362-66-1.
- [12] Piaget, J. (1977). The Development of Thought: The Equilibration of Cognitive Structures. Viking Press.
- [13] Miller, M. S. P. (2013). The Neural Proposition: Structures for Cognitive Systems. AAAI Spring Symposium Technical Report, pp. 44–50.
- [14] Miller, M. S. P. (2026). Coding Artificial Minds. Pre-publication.
- [15] Miller, M. S. P. and Blumberg, M. (2025). Building Sentient Beings. SubThought Corporation. DOI: 10.5281/zenodo.15522356.

Alphabetical order

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Johnson, M. (1987). The Body in the Mind: The Bodily Basis of Meaning, Imagination, and Reason. University of Chicago Press.

Kuhn, T. S. (1962). The Structure of Scientific Revolutions. University of Chicago Press.

Miller, M. S. P. (2013). The Neural Proposition: Structures for Cognitive Systems. AAAI Spring Symposium Technical Report, pp. 44–50.

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Miller, M. S. P. (2026a). Coding Artificial Minds. Pre-publication.

Miller, M. S. P. (2026b). Premise Language Reference Manual, version 3.389. SubThought Corporation.

Miller, M. S. P. and Blumberg, M. (2025). Building Sentient Beings. SubThought Corporation. DOI: 10.5281/zenodo.15522356.

Piaget, J. (1952). The Origins of Intelligence in Children. W. W. Norton and Company. DOI: 10.1037/11494-000.

Piaget, J. (1977). The Development of Thought: The Equilibration of Cognitive Structures. Viking Press.

Tables

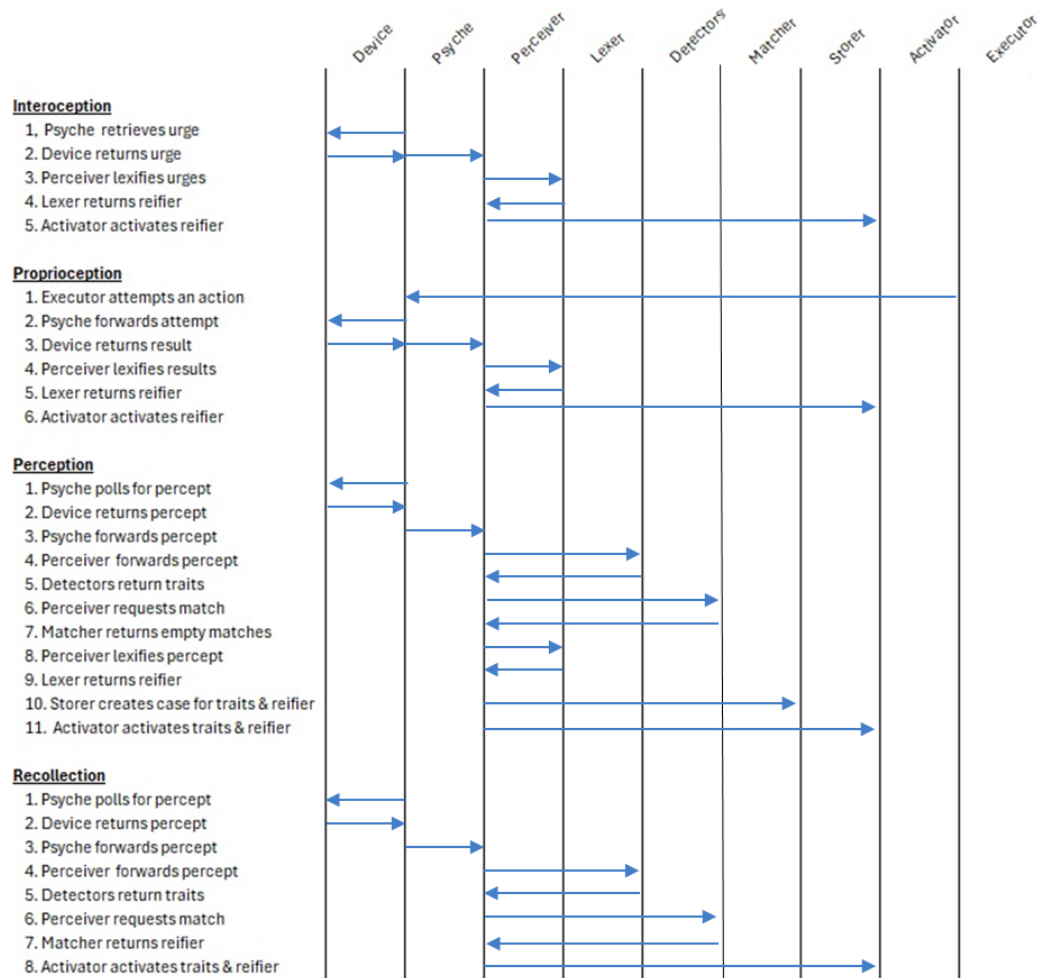


Table 1. Observation Pattern Sequences (simplified).

Score	Mechanism	Meaning
400+p	kace	Direct progress evidence from this exact context (trusted: $4 \times \text{progress} \geq \text{tries}$)
350	dream-progress	Dreamed future is a state progress was made from
340	dream-match	Dreamed action closes a relational match (indicator becomes door glyph)
335-325	chain-progress	Multi-leg walk reaches a progress arrangement (depth 2-4)
332	dream-deliver	Match held, dreamed future carries it strictly closer to the plate
325-315	chain-match	Multi-leg walk closes a relational match (depth 2-4)
300	reactor	Globally reliable action (effects/tries $\geq 80\%$)
250	explorer	Untried anywhere (first trial buys a whole transition model)
250	hold-default	Null action in an untrustable world (distrust ledger crossed 50%)
248	muse-directed	Dreamed future approaches the imported waypoint hypothesis
245	explorer	Untried (arrangement, action) pair (frontier-pair rule)
244-242	chain-carry	Multi-leg walk with match held throughout (depth 2-4)
240-f	dream-novel	Dreamed future is a state never seen (f = curiosity fatigue)
150	residue	Causes effects here, no known progress
120	extrapolated	Untried pair whose action the fallback model predicts is a no-op
108-104	chain-frontier	Multi-leg walk reaches untried pairs (depth 2-4)
105	question-map	Walk proved the map has no way out; re-try the stalest edge
100	dream-known	Dreamed future is a known dead-end
10	dead	Tried here, does nothing
5	inhibited	Model predicts this action does nothing here

Table 2. Band auction hierarchy. Higher scores win; first max breaks ties; candidate order rotates by trial count to prevent persistent ties.

Figures

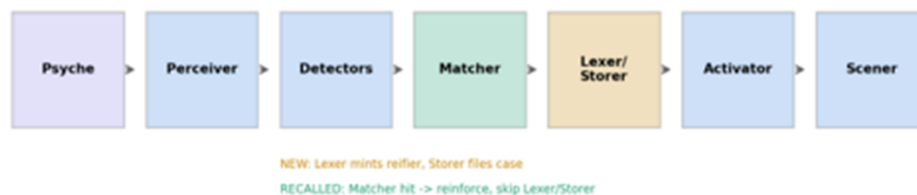


Figure 1. The Perceptual Pipeline.

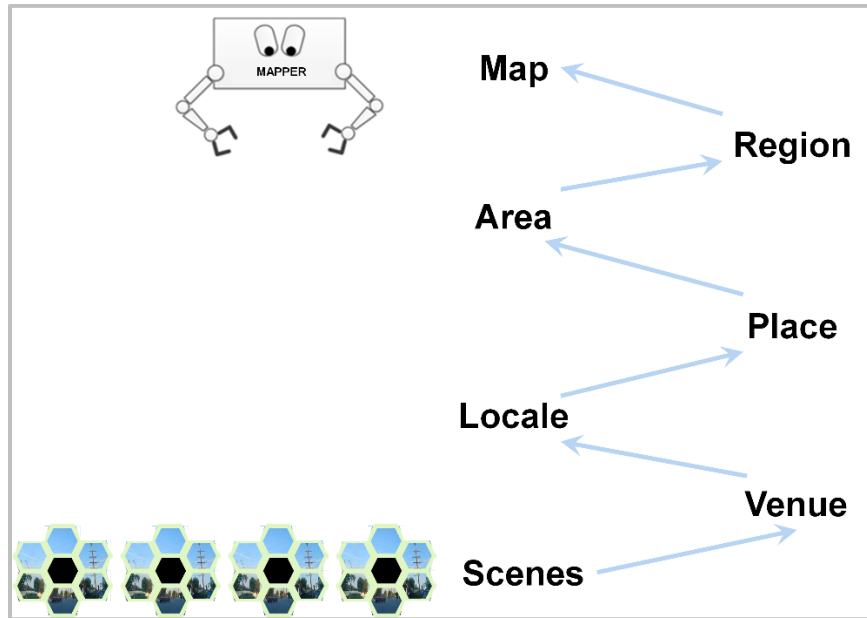


Figure 2. The Mapper accumulates a top down map of the world.

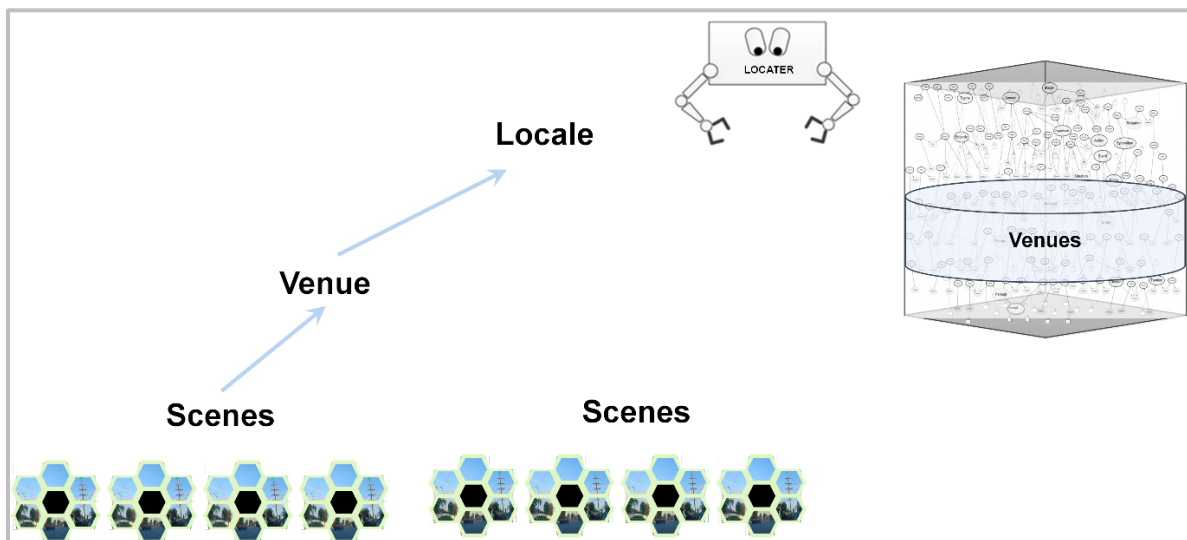


Figure 3. The Locator builds venues bottom up based on perceptions.

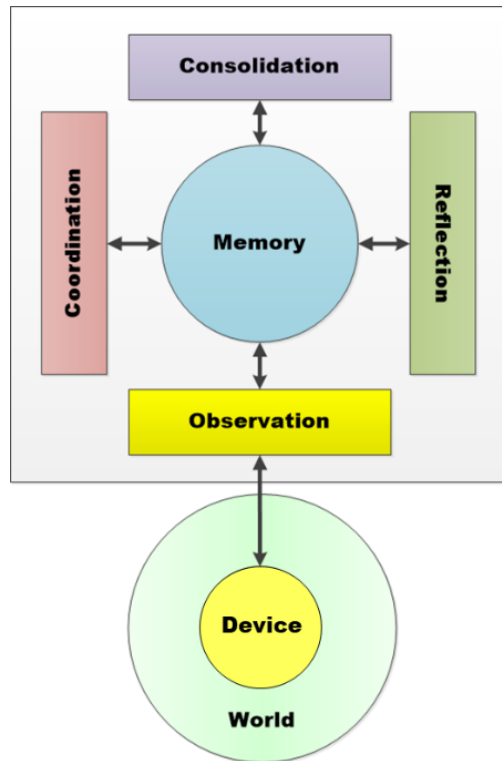


Figure 5. The Piagetian Modeler Cognitive Architecture.

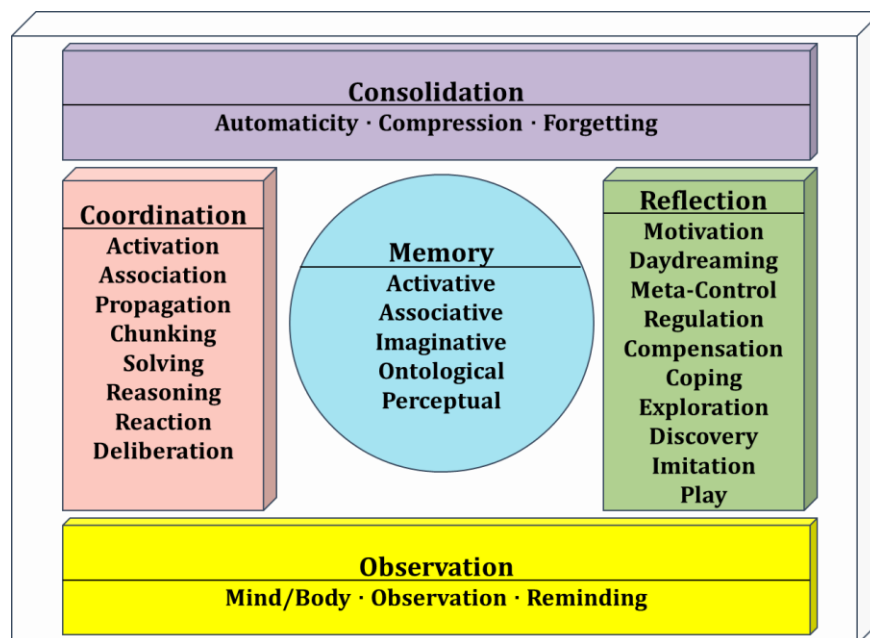


Figure 6. Decomposition of mind processes, design patterns and memory areas in GIL.

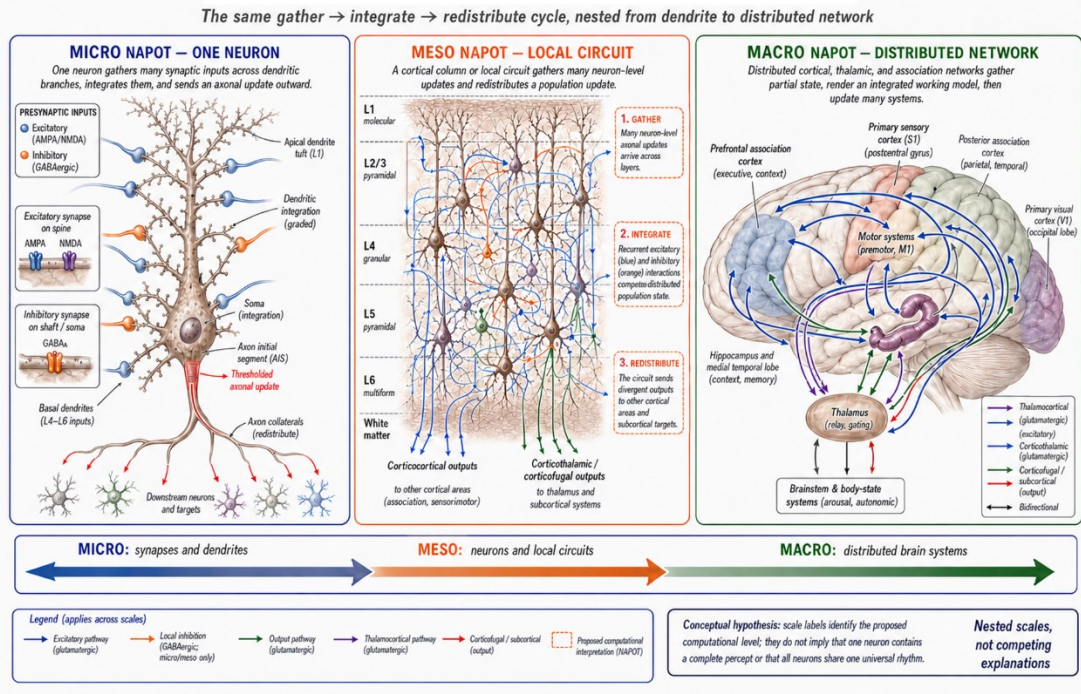


Figure 7. NAPOT at varying levels of description.

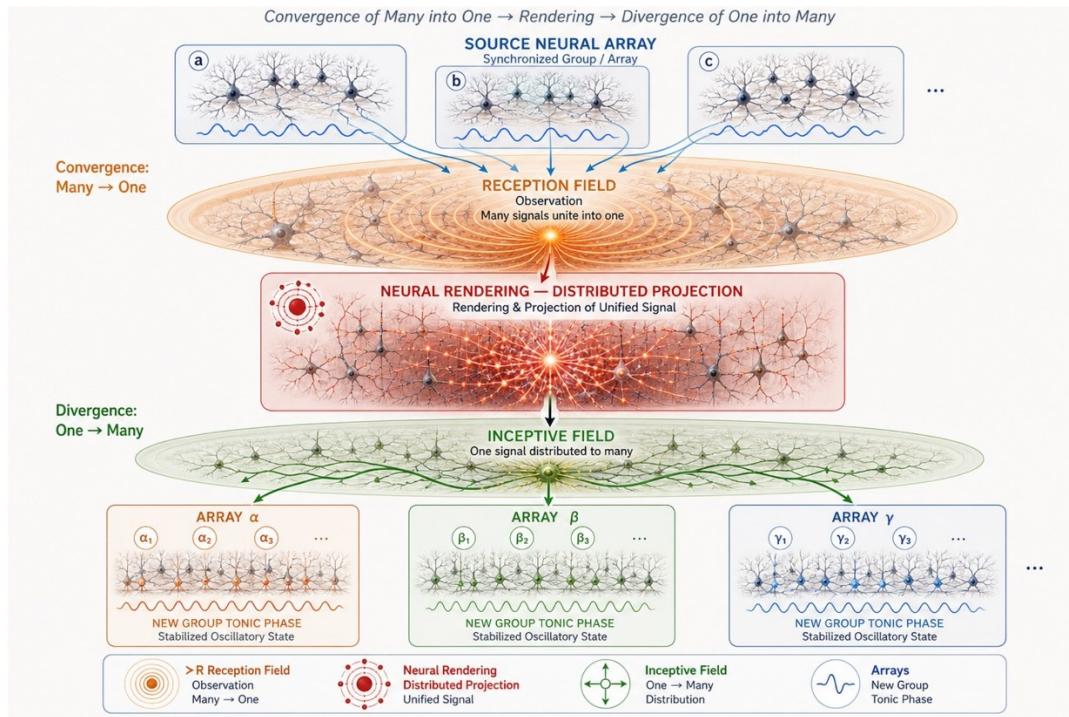


Figure 8. NAPOT principle

```

;;; GIL.being  -- Generally Intelligent Learner

(theory GIL

  (require Observation)
  (require Coordination)
  (require Reflection)
  (require Consolidation)

  (use (knowledge url "locus://sol.earth.orb.local.host/totality")

    (domain Wisdom

      (problem Self-Actualization
        state (assume Novelty :Facts [I_AM] ))

      (concurrent (observe) (coordinate) (reflect)
        (consolidate) (think)))

    ) ;; end use
  ) ;; end theory

;;; ----- end of file -----

```

Figure 9. Source code for GIL.

```

;;; u-GIL.being  -- micro Generally Intelligent Learner

(theory u-GIL

  (grok "./u-GIL.daicho") ;; read settings
  (grok "./mechanisms/u-GIL.mechanisms")
  (grok "./memory/u-GIL.totality")
  (grok "./psyche/camera.psyche")
  (grok "./psyche/grid.psyche")
  (grok "./portal/render.theory")

  (when (= (setting GIL Perceiver :Serve) yes)
    (start (service Perceiver
      (setting GIL Perceiver :Url)
      perceiver-handler
      {perceiving})))

  (when (= (setting Camera Psyche :Serve) yes)
    (start (service Psyche
      (setting Camera Psyche :Url)
      camera-handler
      {perceiving visual})))

  ) ;; end theory

;;; ----- end of file -----

```

Figure 10. Source code for u-GIL.

Charts

The Visible Signature of Learning — Before vs After Mechanism Development

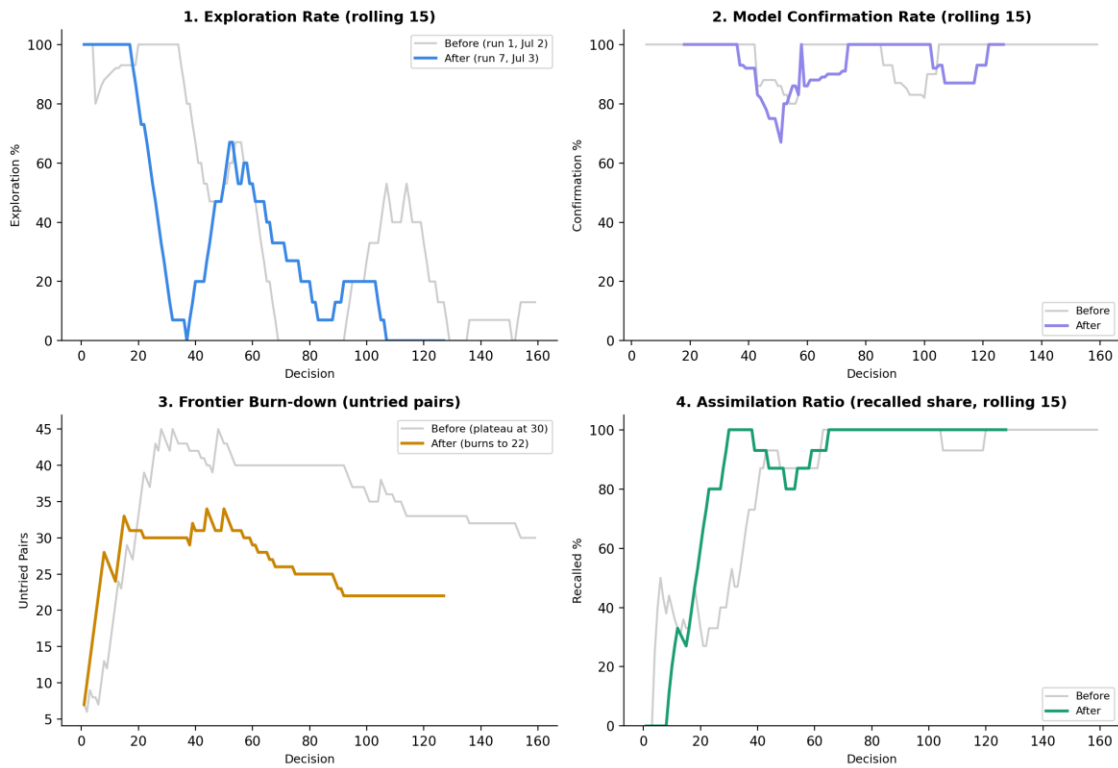


Chart 1. Before vs. After Skill Development.

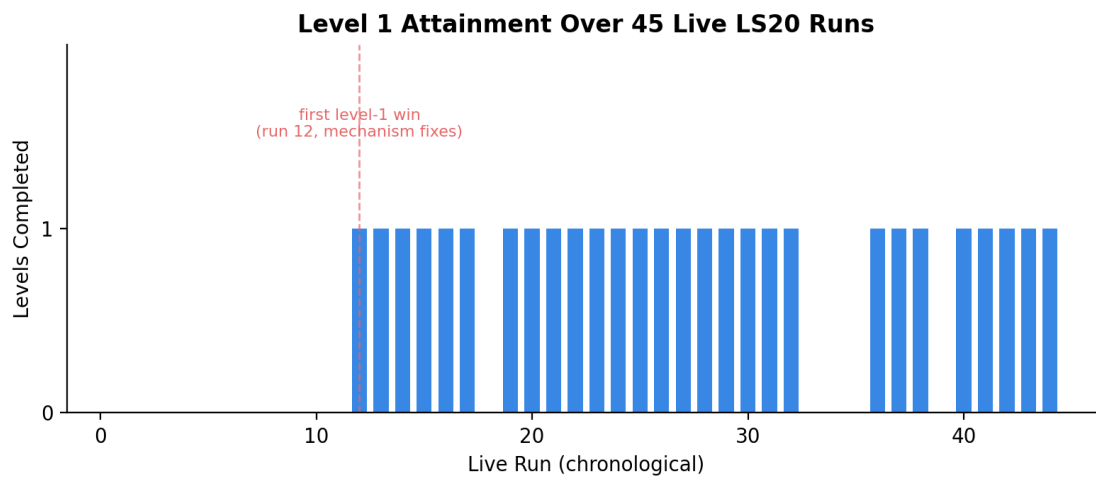


Chart 2. Levels Completed per Run.

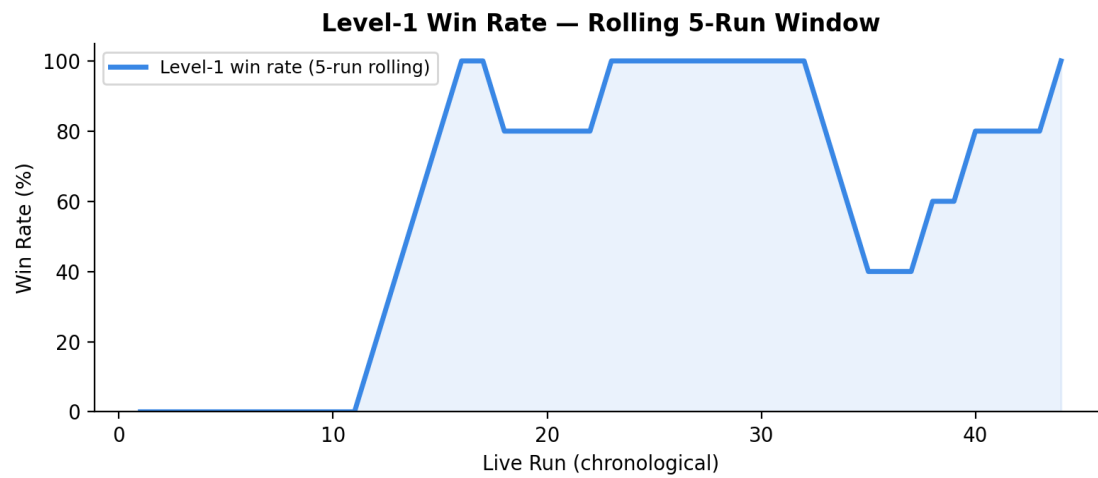


Chart 3. Win Rate

Learning Efficiency — Does the Mind Win Faster?

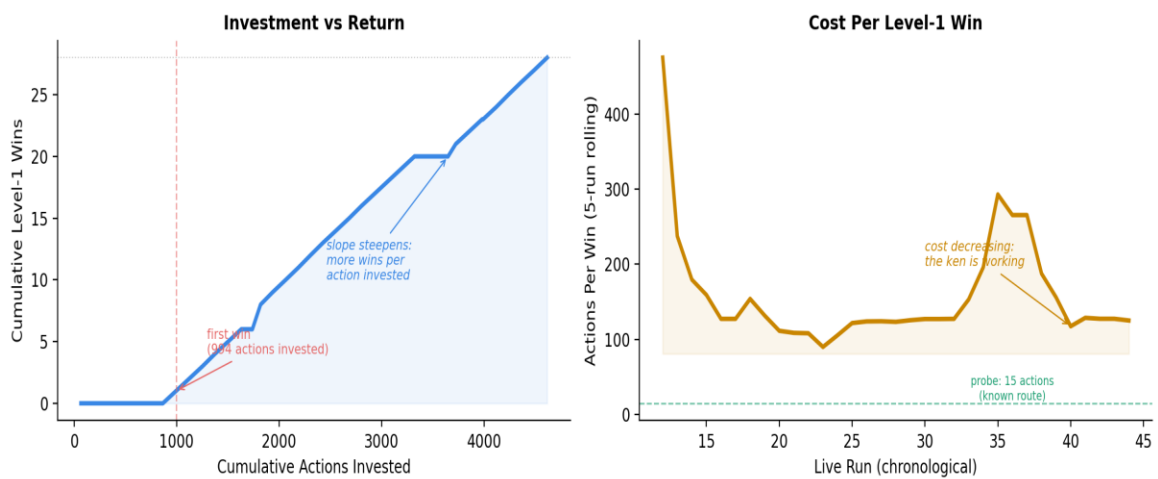


Chart 4. Investment vs. Return, and Cost Per Level-1 Win.